

ClosetScan — App Deep Dive for Interview

AI & AR/VR Innovation Intern · Technical demo prep · Tue Jul 21, 2026 · 9:00 AM EST

What to remember: This is a live RoomPlan + LiDAR demo. Pitch the problem → show the scan → toggle empty shell → read fractional inches → validate vs tape → explain architecture honestly.

1. Problem & product story

Closets are hard to measure by eye, and photos alone do not give usable dimensions for shelving, organizers, or renovation planning. ClosetScan turns a LiDAR iPhone into a closet measuring tool that:

- Scans the physical closet with Apple RoomPlan (LiDAR + cameras).
- Digitally hides contents so reviewers see an empty architectural shell.
- Reports width, depth, and height to the nearest 1/16 inch.
- Lets you validate those numbers against a steel tape and keep a trial history.

Interview framing: Built as a second-round technical demonstration for AI & AR/VR Innovation Intern — not a production App Store product. Focus is reproducible demo, clear architecture, and honest accuracy.

2. Tech stack (say this in 20 seconds)

Layer	Choice	Why
UI	SwiftUI	Fast screens; phase-driven navigation
Scan	RoomPlan + ARKit	Parametric walls/objects from LiDAR
Bridge	UIKit UIViewRepresentable	RoomCaptureView is UIKit
3D	RealityKit (non-AR ARView)	Dollhouse of surfaces ± objects
Math	simd + custom ClosetDimensions	AABB → inches → 1/16"
Persist	UserDefaults / Codable	Validation trial history
Device	iPhone 16 Pro	LiDAR required for RoomPlan
Build	Xcode or GitHub Actions IPA	Sideloadly install path (no Mac required)

3. End-to-end user flow

Home → Scan (RoomCaptureView) → Results (3D + dimensions) → Validation (tape vs app)

Home: Explains the product; checks LiDAR support via

`ARWorldTrackingConfiguration.supportsSceneReconstruction(.mesh)`. Disabled Scan button if unsupported.

Scan: Starts `RoomCaptureSession`. User walks the closet; taps Done → session stops → RoomPlan processes → returns `CapturedRoom`.

Results: RealityKit dollhouse; toggle Show contents; big fractional-inch W/D/H; confidence + surface/object counts.

Validation: Enter tape inches → compute signed error → store trials → show whether $\max |\text{error}| \leq 1/16"$.

4. File-by-file architecture (study this)

ClosetScanApp.swift

App entry. Creates one AppModel as @StateObject and injects it into ContentView.

ContentView.swift / HomeView

Phase switcher (home / scanning / results / validation). Home pitch + Scan Closet CTA.

AppModel.swift

Single source of truth: phase, capturedRoom, dimensions, showContents, scanGeneration, validationStore. Handles beginScan / handleScanComplete / errors.

RoomCaptureController.swift

Owns RoomCaptureView + session start/stop/cancel. Delegate bridge posts roomCaptureDidFinish with CapturedRoom.

ScanView.swift

Embeds RoomCaptureView via UIViewRepresentable. Cancel / Done UI. On finish → AppModel.handleScanComplete.

ClosetDimensions.swift

Math: surface corners → world AABB → width/depth/height → meters, decimal inches, fractional 1/16" display.

EmptyClosetViewer.swift

RealityKit non-AR dollhouse. Always draws walls/floors/openings; objects only if showContents. Pan to orbit.

ResultsView.swift

3D viewer + contents toggle + DimensionsPanel + Validate Accuracy / Scan again.

ValidationView.swift

Tape inputs, error display, trial history, methodology notes.

ValidationStore.swift

Codable trials in UserDefaults; add/clear; computes $|\Delta|$ and $\leq 1/16''$ flag.

5. Scan pipeline (deep dive)

RoomPlan is Apple's high-level API on top of ARKit LiDAR. Instead of giving you only a dense mesh, it returns a **parametric** CapturedRoom:

- **Surfaces:** walls, floors, doors, windows, openings (each has dimensions + transform + confidence).
- **Objects:** furniture / storage bounding boxes (used for the contents toggle).
- Note: RoomPlan has **no ceilings array** in this API surface — height comes from wall extents; viewer synthesizes a thin ceiling lid from floor footprint + wall height.

Lifecycle in code

1. RoomCaptureController.start() → captureSession.run(configuration:)
2. User captures → taps Done → stop() → status = processing
3. Delegate captureView(didPresent:error:) receives CapturedRoom
4. Notification → ScanView → AppModel.handleScanComplete
5. Dimensions computed; phase = results; scanGeneration++ (forces 3D rebuild)

Why a separate DelegateBridge?

RoomCaptureViewDelegate inherits NSCoder requirements. Putting the ObjC-visible delegate on a dedicated NSObject / NSSecureCoding bridge keeps RoomCaptureController (ObservableObject) clean and fixes CI build issues. Good interview story: you hit a real platform constraint and designed around it.

6. Digital empty shell (the demo beat)

Say this: “Digital removal is intentional omission. We render RoomPlan surfaces always, and only add object bounding boxes when Show contents is on. No generative AI inpainting — reproducible under interview lighting.”

In EmptyClosetViewer, each surface becomes a RealityKit box mesh with a SimpleMaterial (walls cool blue-gray, floor green-gray, openings teal translucent, objects orange when shown). Camera is PerspectiveCamera in non-AR mode; UIPanGestureRecognizer orbits the root entity.

7. Dimension math (memorize the steps)

`ClosetDimensions.from(capturedRoom:)` does this:

1. Collect surfaces: walls + floors + doors + windows + openings.
2. For each surface, take 8 local corner points from $\pm \text{dimensions}/2$, transform to world with `surface.transform`.
3. Expand axis-aligned min/max over all world corners (AABB).
4. Fallback: if no samples, use wall centers $\pm$ half dimensions.
5. Extent = max – min. Y $\rightarrow$ height. Larger of $|X|, |Z| \rightarrow$ width; smaller $\rightarrow$ depth (stable labeling).
6. Convert meters $\rightarrow$ inches with $\times 39.37007874015748$.
7. `FractionalInch.format`: round fraction to nearest 1/16, reduce fraction (e.g. 48 3/16").
8. Also report average surface confidence (high/medium/low) and volume in ft^3 .

Why AABB not oriented bounding box? Simpler, stable for interview demo. Trade-off: door frames / openings can slightly expand the box (own this as a known limitation).

8. Validation system

`ValidationTrial` stores app inches + tape inches. Errors are signed (app – tape). `meetsSixteenthTarget` is true when $\max(|\Delta W|, |\Delta D|, |\Delta H|) \leq 0.0625"$. History persists via JSON in `UserDefaults` so you can show multiple practice trials live.

Protocol to run before interview (and show):

1. Open closet door fully; good light; clear walking path.
2. Scan slowly — every wall, floor corners, upper extent.
3. Record app W/D/H from Results.
4. Measure same reference edges with steel tape (decimal inches).
5. Enter tape values $\rightarrow$ Compute error.
6. Repeat ≥ 3 independent scans. Speak MAE / max error, not marketing claims.

Accuracy honesty line (must say): Display resolution is 1/16". LiDAR + RoomPlan is typically centimeter-class. We do not claim guaranteed $\pm 1/16"$ sensor accuracy — we measure, report to 1/16", and validate against tape.

9. Device, permissions, constraints

- Requires LiDAR (checked at runtime). iPhone 16 Pro is supported. Simulator cannot run RoomPlan capture.
- `Info.plist`: `NSCameraUsageDescription` + `NSWorldSensingUsageDescription`; `UIRequiredDeviceCapabilities` includes `arkit` + `metal`.
- Portrait phone orientation for the demo.
- Free install path: GitHub Actions builds IPA $\rightarrow$ Sideloadly on Windows $\rightarrow$ free Apple ID (7-day re-sign). Re-sideload before interview if needed.

10. Known limitations (own them calmly)

- Non-LiDAR phones cannot capture.
- Narrow / cluttered closets may miss walls or merge surfaces.
- Hanging clothes may detect inconsistently as objects; empty-shell still shows structure.
- AABB can overestimate when openings expand bounds.
- Width vs depth labeling is normalized (larger horizontal = width).
- Synthesized ceiling is approximate (no RoomPlan ceilings API).

11. Strong Q&A answers

Q: Walk me through the architecture.

A: SwiftUI phase machine in AppModel. Scan uses RoomCaptureView; results use RealityKit dollhouse from CapturedRoom surfaces; ClosetDimensions computes AABB; ValidationStore persists tape comparisons.

Q: How do you digitally empty the closet?

A: Omission, not inpainting. Surfaces always drawn; objects drawn only when toggle is on. That is the before/after beat.

Q: How do you get 1/16 inch?

A: Convert meters to inches, round the fractional part to nearest sixteenth, reduce the fraction for display. Validation checks absolute error against 0.0625".

Q: Is RoomPlan accurate enough?

A: Honestly centimeter-class. Fractional inches are reporting precision. We prove quality with tape trials, not claims.

Q: What would you ship next?

A: Opening-aware clear span (exclude door swing), multi-scan fusion, USDZ export, better ceiling estimate, charts for validation MAE, optional ML clothing segmentation on top of the shell.

Q: Biggest engineering snag?

A: RoomPlan API quirks (no ceilings array) and RoomCaptureViewDelegate / NSCoder for CI. Separated a DelegateBridge NSObject to keep ObservableObject clean.

Q: Why not raw ARKit mesh?

A: Parametric surfaces + objects give a cleaner empty-shell story and stabler dimensions for a short demo than fighting a dense mesh.

12. Condensed 20-minute script

0:00–0:30 — Home pitch: RoomPlan + LiDAR + RealityKit; empty shell + 1/16" + tape validation.

0:30–3:30 — Live scan — narrate covering walls/corners/top.

3:30–5:00 — Toggle contents On → Off (empty shell beat).

5:00–6:00 — Read W×D×H fractional inches + confidence.

6:00–8:00 — Validation: enter tape, show |error| and history.

8:00–16:00 — Architecture, math, honesty, limitations, next steps.

16:00–20:00 — Q&A.;

13. Closing line

"Thanks for the time — happy to walk through any part of the scan pipeline, the dimension math, or the validation results in more detail."